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30 September 1977
SUPERSEDING
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23 April 1973

MILITARY SPECIFICATION

FORCEPS, HEMOSTATIC, CURVED, MAYO-CARMALT, 8 INCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a curved hemostatic forceps, of the Mayo-Carmalt design, suitable for use in general surgical procedures.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-Pakaging, Methods of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

ASTM E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.

ASTM E 92 - Vickers Hardness of Metallic Materials, Tests for.

ASTM E 140 - Standard Hardness Conversion Tables for Metals (Relationship Between Brinell Hardness, Vickers Hardness, Rockwell Hardness, Rockwell Superficial Hardness and Knoop Hardness).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference form libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 General requirements. The forceps shall be in accordance with the applicable requirements of GG-I-526 and as specified herein.

3.2 Material. The forceps shall be fabricated from Class 4, type 410 or 420 material.

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3.3 Hardness. After proper stress relieving, the forceps shall have a Rockwell hardness of C40 to C45 when tested as specified in 4.4.1. The requirements of GG-I-526 shall apply.

3.4 Style, design and dimensions. Shall be a curved hemostatic forceps of the Mayo-Carmalt design. The forceps shall be suitable for accomplishing hemostasis in delicate tissues during general surgical procedures. Both jaws of the forceps shall be serrated as shown in figure 1. Serrations shall be consistent in depth and in accordance with GG-I-526. Jaw serrations shall be chamfered throughout their full length. The forceps shall mesh completely and properly during closure. The forceps shall be in strict accordance with figure 1.

3.5 Locks. Shall be of the box lock type. The box lock shall be accurately fitted, without crevices, burrs, or sharp edges at any place in the lock construction. Pin or rivet shall be permanently secured. The joint performance shall be smooth, of equal resistance, and non-binding in all positions when opening or closing the forceps (the lock may be lubricated). Further, under no circumstances shall looseness or free play or either shank permit any misalignment of the jaws in closing. NOTE: This paragraph shall replace the requirements specified in GG-I-526 for Locks.

3.6 Finish. Finish shall be entirely mirror, satin or matte. The locks and ratchets of instruments with a mirror finish shall be polished smooth, but need not possess a mirror finish. The matte finish shall exhibit a uniform nondirectional surface texture equivalent to that produced by lightly blasting the prepared surface with fine abrasives under pressure.

3.6.1 Passivation. The forceps shall be properly passivated.

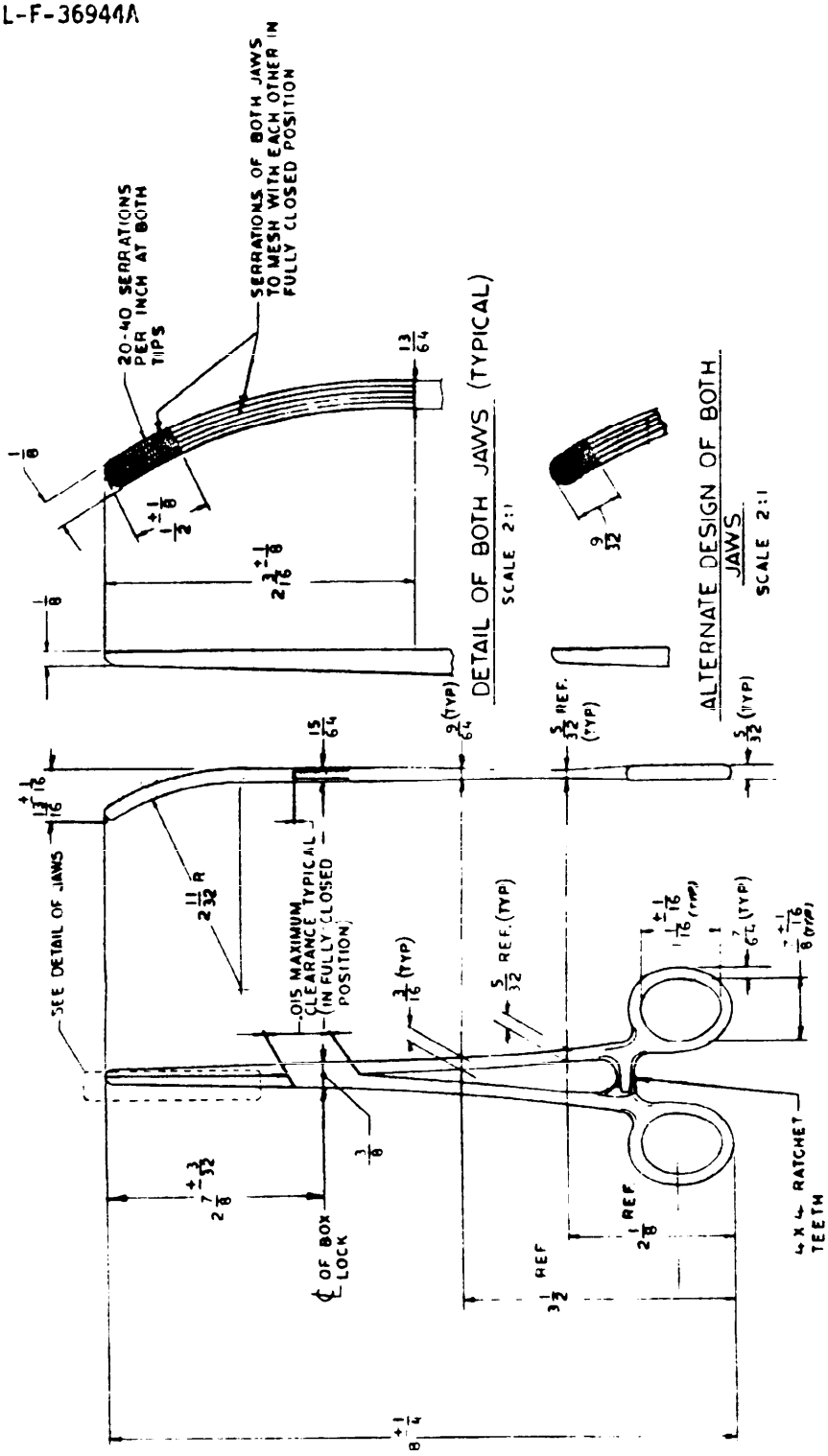
3.7 Jaw and ratchet operation. When tested as specified in 4.4.2, the jaw tips shall close first and progressively tighten with each ratchet position. Each ratchet position shall hold firmly and release instantly.

3.8 Identification marking. In addition to GG-I-526 requirements, forceps of foreign manufacture shall also be marked "stainless".

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance if the inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or

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NOTE:
TOLERANCES INDICATED DELINEATE ACCEPTABLE EXTREME DIMENSIONS AND ARE NOT INTENDED TO PERMIT VARIATIONS WITHIN A SINGLE INSTRUMENT WHICH WILL SACRIFICE UNIFORMITY, SYMMETRY, AND FIRST QUALITY WORKMANSHIP

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES AND TOLERANCE IS $\pm\frac{1}{32}$ INCH

FIGURE 1. Forceps, hemostatic, curved, Mayo-Carmalt, 8 inch.

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any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection shall be in accordance with GG-I-526.

4.1.2 Records. Records of examination and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the corrosion resisting steel may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificates shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, the individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table I. The unit of product for examination purposes shall be one forceps.

TABLE I. Sampling criteria for examination.

Type of defect	Inspection level	AQL (percent defective)
Major	I	1.0
Minor	II	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product for test purposes shall be one forceps. Where feasible, the sample unit shall be used for the determination of test characteristics.

TABLE II. Sampling criteria for tests.

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Forceps	Hardness	3.3 and GG-I-526	4.4.1	S-2	1.5
	Dimensions	3.4	-----	S-2	2.5
	Boil test	GG-I-526	GG-I-526	S-2	1.0
	Copper sulfate	GG-I-526	GG-I-526	S-2	1.0
	Distortion test	GG-I-526	GG-I-526	S-2	2.5
	Jaw and ratchet operation test	3.7	4.4.2	S-2	2.5

4.2.3 Raw material. A sufficient quantity of each lot of metal used in the manufacture of the forceps shall be selected for chemical analysis.

4.3 Examination. The forceps shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any forceps in the sample containing one or more defects shall be rejected, and if the number of defective items in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

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TABLE III. Classification of defects.

Categories	Defects
Critical	None defined.
Major	
101	Forceps not of the design specified.
102	Jaw serrations not of uniform depth or not well defined.
103	Jaw serrations not chamfered throughout full length.
104	Ratchet teeth not well defined.
105	Forceps not free from cracks, pits, burrs, sharp edges, scale or corrosion.
106	Finish of forceps not uniform and not as specified.
107	Box lock performance not as specified in 3.5.
108	Serrations of both jaws do not completely mesh in fully closed position.
109	Jaws possess sharp tips.
110	Ratchet teeth do not completely engage in all ratchet positions.
111	Pin for box lock not flush with surface and not free from crevices.
Minor	
201	Instruments not free from dirt, polishing material or other foreign matter.
202	Identification markings not present, not correct, not complete, or not legible.

4.3.2 Packaging inspection. The inspection of the preservation-packaging shall be in accordance with the Group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Hardness test. Hardness shall be determined in accordance with ASTM Std. Method E 1B or E 92. Conversion of Vickers hardness values to Rockwell values shall be in accordance with ASTM Std. Hardness Conversion Tables E 140.

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4.4.3 Jaw and operation test. The forceps shall be positioned to the first ratchet position, but not engaged. Forceps with jaw tips not touching shall be rejected; further, should all the jaw serrations touch, the forceps shall be rejected. The forceps shall then be engaged to each of the ratchet positions. Forceps that do not progessively tighten with each ratchet advance shall be rejected.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Each forceps shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.1.2 Preservative. Preservative shall not be applied.

5.1.1.3 Unit packaging. Each forceps shall be wrapped in neutral tissue or other suitable anticorrosive materials. Pliers shall be further packaged in accordance with MIL-P-116, method IC-1.

5.1.1.4 Intermediate packaging. Six (6) forceps shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-566, PPP-B-636, type CF, class domestic or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.2 Level C. Forceps shall be clean and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level a, B or C as specified (see 6.1).

5.2.1 Level A. Two hundred eighty-eight (288) holders shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, type CF, class weather resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

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5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636.

5.2.2 Level B. Two hundred eighty-eight (288) holders shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636 type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The packaged forceps shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking. Each unit, intermediate package, and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Number, title and date of this specification.
(b) Selection of applicable levels of packaging required (see 5.1 and 5.2).
(c) NSN.

6.2 NSN coverage. This specification covers the following item appearing in the Federal Supply Catalog:

NSN 6515-00-334-4100 FORCEPS, HEMOSTATIC, Curved, Mayo-Carmalt, 8 inch.

6.3 Matte and satin finish. It should be noted that matte finish specified herein can be distinguished from mirror and satin finishes under high-power illuminated magnification. Forceps of matte finish shall have a reduced reflective surface that is more glare resistant than satin finish.

6.4 Changes from previous issues. Asterisks to indicate changes have not been used in this specification due to the extensiveness of the changes.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-36944A FORCEPS, HEMOSTATIC, CURVED, MAYO-CARMALT, 8 inch

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